



TB AUDIO PLUG-INS
Team Impulse Impact

USER MANUAL

TB Pitch Shifter

Pitch and Formant Shifter

Moves pitch across four octaves and the formant across two, independently, with real-time FFT processing and a fixed, reported latency.

Pitch ± 24 semitones

Formant ± 12 semitones

Two main controls

5 factory presets



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01 Overview

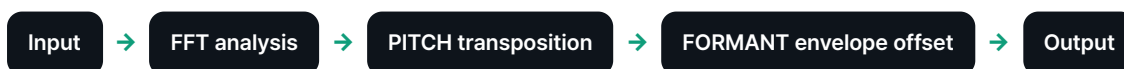
TB Pitch Shifter is a focused tool for moving pitch and formant separately. **PITCH** shifts the harmonic spacing from -24 to +24 semitones. **FORMANT** moves the perceived size of a voice or instrument across its own range of -12 to +12 semitones, without forcing the note to follow.

With only two main controls you can make clean octave changes, bigger or smaller characters, transformed effects, or tonal reshaping that speed-based playback cannot achieve. Processing runs in real time in mono or stereo, with smooth automation, a fixed reported latency, and an aligned bypass. With both controls at 0 st, the output returns exactly to the dry signal.

Key features

- **PITCH** from -24 to +24 semitones in 0.01-semitone steps
- **FORMANT** from -12 to +12 semitones, independent of the note
- Real-time FFT processing for mono and stereo
- Fixed reported latency and latency-aligned bypass
- Exact dry output at 0 st on both controls
- 5 factory presets, user presets, undo, and redo

Signal flow



About this manual

This manual describes TB Pitch Shifter 2.0.0 for Windows (VST3). The current macOS build is version 1.0.2 and may differ from the descriptions here.

02 Quick start

- 1 Insert TB Pitch Shifter on a vocal or instrument track. Both controls start at 0 st, which leaves the sound unchanged.
- 2 Turn **PITCH** to transpose: +12 st is an octave up, -12 st an octave down.
- 3 Turn **FORMANT** to change the apparent size: negative values sound larger and darker, positive values smaller and brighter.

- 4 Use the Female to Male or Male to Female presets as starting points for voice changes.

03 Interface



TB Pitch Shifter 2.0 interface (portfolio image).

Area	What it does
Preset bar	Previous and next preset, the preset browser (factory and saved user presets; Custom after a change), Save, Undo, and Redo.
PITCH dial	Transposition, ± 24 semitones.
FORMANT dial	Formant offset, ± 12 semitones.

04 Controls

Control	Range · default	What it does
PITCH	-24 to +24 st · 0 st	Moves the pitch with natural spectral and formant movement. Large shifts make processing artifacts more audible.
FORMANT	-12 to +12 st · 0 st	Offsets the spectral envelope relative to the transposed signal. Negative values sound larger and darker; positive values smaller and brighter. Extreme values sound unnatural.

TB Pitch Shifter also has a host Bypass parameter. The plug-in reports a fixed latency to the host, and the bypassed signal stays time-aligned with the processed one.

To change a voice's character without changing the note, leave PITCH at 0 st and move only FORMANT.

05 Presets

Step through presets with the arrows, or open the preset browser to choose a factory or saved user preset. Save stores the current settings as a .tbps file. Undo and Redo step through recent parameter and preset changes.

Group	Preset	PITCH	FORMANT
Start	Init	0 st	0 st
Pitch	Octave Up	+12 st	0 st
Pitch	Octave Down	-12 st	0 st
Voice	Female to Male	-4 st	-1.5 st
Voice	Male to Female	+4 st	+1.5 st

06 Recipes

Starting points for common jobs. Adjust them by ear.

Octave layer

A doubled octave under or over a lead.

Preset Octave Up or Octave Down

Use On a duplicate track

- 1 Blend the shifted track under the original with the mixer fader.

Voice change

A different-sounding speaker or singer.

Preset Female to Male or Male to Female

FORMANT Fine-tune by ear

- 1 Small formant changes are often more convincing than large pitch changes.

Bigger or smaller character

A creature, robot, or cartoon voice.

PITCH 0 st or a small shift

FORMANT -6 to -12 st (big) or +6 to +12 st (small)

- 1 Keep PITCH near 0 so the words stay on the original notes.

Instrument tone reshaping

Darker or brighter body on guitar or synth.

PITCH 0 st

FORMANT -3 to +3 st

- 1 Compare with bypass at matched loudness.

07 Tips and troubleshooting

Symptom	What to check
The shifted sound is smeared or phasey.	Large shifts make FFT artifacts more audible. Use smaller shifts, or blend the result on a parallel track.
The voice sounds like a chipmunk or a giant.	Adjust FORMANT to bring the apparent size back toward natural.

Symptom	What to check
The track is late against others.	Make sure the host's latency compensation is on; the plug-in reports its fixed latency.
Nothing changes.	Both controls are at 0 st, or the host Bypass parameter is on.

08 Parameter reference

Panel label	Host parameter	Range	Default
PITCH	Pitch	-24 to +24 st	0 st
FORMANT	Formant	-12 to +12 st	0 st
—	Bypass	Off, On	Off

09 Version history

Version	Changes
2.0.0 · 2026-08-14	Lower-hop STFT processing, smoother formant automation, and an exact dry signal at the neutral setting. The editor was redesigned as a bright dual-instrument panel; Pitch and Formant stay independent.
1.0.0 · 2026-08-02	Independent Pitch (± 24 st) and Formant (± 12 st), five factory presets, and user preset saving and loading.

10 Support and license

Report problems or request features at github.com/TaebeumKim/TB_Audio-Plugins/issues. Include the plug-in version, your DAW, and your operating system.

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